Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Overview
This project uses Machine Learning techniques to analyze real estate buyer behavior and segment customers based on demographic, financial, and investment characteristics. The project helps identify different buyer groups, enabling better marketing strategies and data-driven investment decisions.

Objectives
- Clean and preprocess the datasets
- Perform Exploratory Data Analysis (EDA)
- Segment buyers using K-Means Clustering
- Analyze investment behavior
- Build an interactive Streamlit dashboard
- Generate business insights and recommendations

Dataset
The project uses two datasets:
- **clients.csv** – Client demographic and investment information
- **properties.csv** – Property details and purchase information

Technologies Used
- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Plotly
- Streamlit
- Jupyter Notebook
- GitHub

Project Structure
```
RealEstate-Buyer-Segmentation/
├── README.md
├── requirements.txt
├── app.py
├── clients.csv
├── properties.csv
├── data_preprocessing.ipynb
├── buyer_segmentation_model.ipynb
├── kmeans_clustering.ipynb
├── project_report.pdf
├── project_presentation.pptx
```

Machine Learning Workflow
1. Data Collection
2. Data Cleaning
3. Data Preprocessing
4. Feature Engineering
5. Feature Scaling
6. K-Means Clustering
7. Cluster Evaluation
8. Business Insights

9. Interactive Streamlit Dashboard

Dashboard Features
- Buyer Segmentation Overview
- Investor Behavior Analysis
- Geographic Buyer Analysis
- Property Analysis
- Cluster Distribution
- Interactive Filters
- Business Insights

Results
The project successfully segments buyers into meaningful groups using K-Means clustering and provides valuable insights into customer behavior, investment patterns, and property preferences.

Future Scope
- Advanced Machine Learning models
- Recommendation System
- Real-time Analytics
- Cloud Deployment